Assign 3 bda

Delete the failed output folder:

Bash

```
hadoop fs -rm -r /grading_output
```

1. The Logic

- **Input:** A text file with StudentName,Marks.
- **Mapper:** Reads the marks, applies an if-else condition to determine the grade, and outputs (Grade, StudentName).
- **Reducer:** Groups all students who received the same grade together.

---

2. The Input File (students.txt)

Create this in your ~/mapreduce folder:

Plaintext

```
Shreya,95

Amit,82

Sneha,74

John,61

Doe,45
```

---

3. The Simple Python Implementation

mapper.py

Python

```
#!/usr/bin/env python3

import sys


for line in sys.stdin:
```

```
    line = line.strip()
    if not line: continue

    # Split the name and marks
    name, marks = line.split(',')
    marks = int(marks)

    # Determine Grade
    if marks >= 90: grade = "A+"
    elif marks >= 80: grade = "A"
    elif marks >= 70: grade = "B"
    elif marks >= 60: grade = "C"
    else: grade = "F"

    # Output: Grade as Key, Name as Value
    print(f"{grade}\t{name}")
```

reducer.py

```
#!/usr/bin/env python3
import sys
for line in sys.stdin:
    line = line.strip()
    if not line or ',' not in line:
        continue
    try:
        # rsplit(',', 1) splits only at the LAST comma
        name, marks = line.rsplit(',', 1)
```

```
        marks = int(marks.strip())
        # Grading Logic
        if marks >= 90: grade = "A+"
        elif marks >= 80: grade = "A"
        elif marks >= 70: grade = "B"
        elif marks >= 60: grade = "C"
        else: grade = "F"
        print(f"{grade}\t{name}")
    except ValueError:
        # This skips lines that don't have a valid number at the end
        Continue
# #!/usr/bin/env python3
# import sys

# for line in sys.stdin:
#     line = line.strip()
#     if not line: continue

#     # Split the name and marks
#     name, marks = line.split(',')
#     marks = int(marks)

#     # Determine Grade
#     if marks >= 90: grade = "A+"
#     elif marks >= 80: grade = "A"
#     elif marks >= 70: grade = "B"
```

elif marks >= 60: grade = "C"

else: grade = "F"

Output: Grade as Key, Name as Value

print(f"{grade}\t{name}")

---

4. Exam Execution Steps

1. **Permissions:** chmod +x mapper.py reducer.py
2. **HDFS Prep:**

Bash

```bash
hadoop fs -rm -r /grading_output

hadoop fs -mkdir -p /input

hadoop fs -put students.txt /input/
```

3. **Run the Job:**

Bash

```bash
hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \

-file mapper.py -mapper "python3 mapper.py" \

-file reducer.py -reducer "python3 reducer.py" \

-input /input/students.txt \

-output /grading_output
```

4. **Check Result:** hadoop fs -cat /grading_output/part-00000